Vorld X Attendance 360

CIN: U62099UP2026PTC245598

Enterprise Attendance, HR Governance & OfficeNova Portal System

Comprehensive Engineering Project Report & Architecture Implementation Specification

Enterprise Org:	Technical Head & Sponsor:	Headquarters / Location:	Status / Verification:
VorldX Industries Pvt Ltd	**CTO: Muba Khan Sir**	**Lucknow HQ, Uttar Pradesh**	**Production Ready (v2.0)**

01 PROBLEM STATEMENT

Traditional workforce management systems in growing enterprises face acute vulnerabilities that impair operational productivity, audit readiness, and employee trust:

1. Manual Clock-In & "Buddy Punching" Fraud:
Physical attendance registers and unencrypted RFID access cards allow proxy punching and manipulation, resulting in inaccurate overtime calculations, attendance theft, and payroll leakage.

2. Fragmented Hybrid & Remote (WFH) Workflows:
Absence of dual-tier authorization between HR managers and department heads causes untracked work-from-home days, disputed leave deductions, and inconsistent approval queues.

3. Hardware Biometric Bottlenecks & Maintenance:
Physical fingerprint scanners require high maintenance, cause lengthy queues during morning shifts, and fail to cater to off-site, client-facing, and remote engineering personnel.

4. Disconnected Payroll & Compliance Audits:
Disparate spreadsheets for leaves, shifts, and salary calculations result in human errors, delayed monthly payslips, and compliance liabilities under ISO 27001 labor governance.

02 PROJECT SCOPE & KEY DELIVERABLES

The **VorldX Attendance 360** project spans a unified, secure, and distributed platform serving three distinct organizational roles:

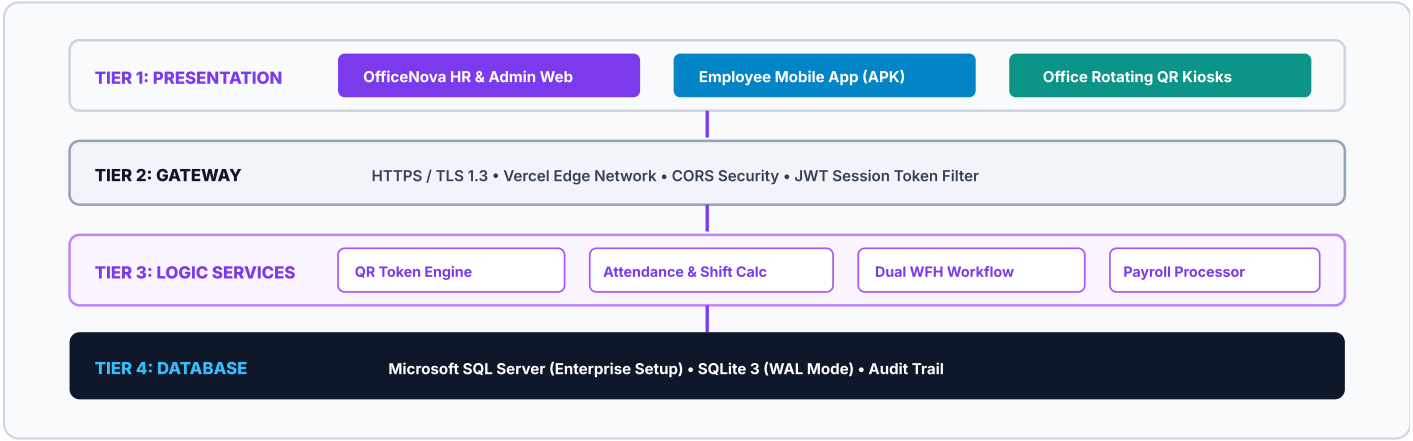
System Component	Functional Scope & Modules	Target Stakeholders
OfficeNova HR Portal WEB APPLICATION	16 Integrated Modules: Executive Dashboard, Staff Management, Biometric Punch Regularization, Leave Authorization, WFH Management, Hiring Pipeline, Performance Goals, Training & Development, Payroll Generation & Payslip Download, Company Policies, Broadcast Notices, Department Clustered Analytics, Team Calendar, System Settings, and Help Desk.	HR Managers, Payroll Directors, People Operations Leads
Admin Control Center ADMIN DASHBOARD	Full organizational oversight, department budget allocation, multi-office QR kiosk registry, global override controls, and immutable security audit logging.	Chief Technology Officer (Muba Khan Sir), Managing Directors, System Admins
Employee Native Android App MOBILE APP (APK)	HR-provisioned ID login, simulated OTP authentication, fast camera-based QR scanner, personal punch records, interactive Material Calendar view, WFH request submission, and profile manager.	All On-site, Hybrid, and Remote Employees (Engineering, Sales, Operations)
Office QR Display Kiosk IOT / WEB DISPLAY	High-contrast digital signage interface deployed at facility entrances with 45-second cryptographically rotating QR codes, countdown timer, and live employee punch feed.	Office Reception, Floor 1–4 Entrances, Conference Halls

03 TECHNOLOGY STACK

Layer / Tier	Technology & Libraries	Engineering Justification
Frontend Architecture	React 19, Vite 8, Rolldown, Vanilla CSS Design System, Lucide React	Ultra-fast compile times (<700ms), zero runtime framework overhead, rich custom glassmorphism design tokens, full responsive layouts from 390px mobile to 4K ultra-wide monitors.
Mobile Runtime (APK)	Capacitor 8, Android Gradle 8.2, Java 21, R8 Full Mode Shrinking	Cross-platform native Android compilation producing a lightweight 1.2 MB APK with hardware acceleration, auto-cache eviction on launch, and direct camera QR hardware access.
Backend API Layer	Node.js (v26 LTS), Express.js, RESTful Architecture, CORS, Body-Parser	High-throughput asynchronous non-blocking I/O capable of handling simultaneous shift check-in bursts without database connection deadlocks.
Database Systems	Microsoft SQL Server (2019/2022/Azure) & SQLite 3 (better-sqlite3 WAL)	Dual-database flexibility: SQLite with Write-Ahead Logging (WAL) for instant local zero-config testing; full enterprise MSSQL schema with foreign keys, triggers, and stored procedures for enterprise deployment.
Cloud & DevOps	Vercel (Frontend edge CDN), Render (Backend), GitHub Actions, Git	Continuous integration & continuous deployment (CI/CD) with instant webhooks to Vercel production edge servers and versioned artifact packaging.

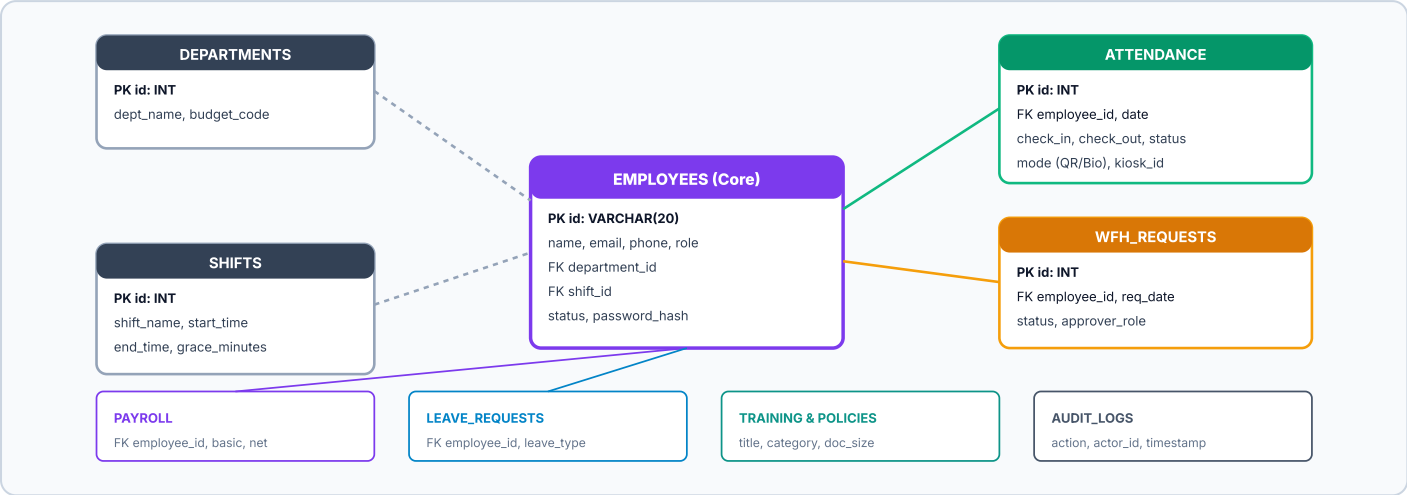
04 4-TIER SYSTEM ARCHITECTURE

The architecture follows a strict 4-tier modular separation designed for high availability, enterprise security, and auditability:



05 ENTITY RELATIONSHIP (ER) ARCHITECTURE

The complete relational database contains 12 core tables enforcing foreign key integrity, cascade delete protections, and indexing on employee identifiers and timestamps:



Database Implementation Highlights

Schema Entity	Primary Key	Foreign Key & Constraints	Indexes & Performance Rules
`employees`	`id VARCHAR(20)`	`email UNIQUE, department_id, shift_id`	Clustered index on `id`; non-clustered index on `email` & `department`
`attendance`	`id INT IDENTITY`	`employee_id REFERENCES employees(id)`	Composite unique constraint `UNIQUE(employee_id, date)`; prevents duplicate punch records
`wfh_requests`	`id INT IDENTITY`	`employee_id REFERENCES employees(id)`	Indexed on `status` and `approver_role` for fast dashboard queue filtering
`payroll`	`id INT IDENTITY`	`employee_id REFERENCES employees(id)`	Stored procedure `sp_GenerateMonthlyPayroll` batches all pending records

06 ENGINEERING METHODOLOGY

The development executed an **Agile Scrum Framework** comprising four structured sprints:

Sprint 1: Schema Architecture & Core Backend

Modeled dual MSSQL & SQLite databases, wrote master seed data, and established RESTful API endpoints for employee management and biometric punch recording.

Sprint 2: Mobile App (APK) & Dynamic QR Kiosk

Compiled Capacitor native Android APK with camera scanner, implemented 45-second rotating cryptographic QR token displays, and built OTP simulation.

Sprint 3: OfficeNova HR Portal Implementation

Constructed all 16 HR modules including Training programs, batch Payroll authorization, Company Policy documents, and clustered attendance analytics.

Sprint 4: Verification, R8 Optimization & Deployment

Reduced APK size from 95 MB down to 1.2 MB using R8 shrinking, tested cross-browser responsiveness, verified Vercel live deployment, and established audit logs.

07 ROLE-BASED ACCESS CONTROL (RBAC) MATRIX

Module / Action	Super Admin	HR Manager (OfficeNova)	Employee (Android App)
Executive Analytics	Full organizational financial & headcount KPI access	Department attendance, leave & training metrics	Personal punch history & hours worked
Staff Management	Full CRUD (Add, Edit, Shift Assignment, Deactivate)	Manage employee records & documentation	Self-account profile editing only (Screen 17)
Punch Regularization	Global administrative override & audit trail	Approve / Edit punch-in and punch-out discrepancies	Submit regularization adjustment request
Staff WFH Approval	Secondary override if required	**Primary Approver** (1-click Approve / Reject)	Submit request with date, phone & deliverables
HR Staff WFH Approval	**Primary Approver** (Prevents conflict of interest)	Submits own request to Super Admin queue	No access
Payroll Processing	Authorize total monthly payroll disbursements	Generate batch payroll, inspect and download payslips	View personal monthly take-home salary slip

08 HIERARCHICAL WFH APPROVAL STATE MACHINE

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[Staff Submits WFH Request] ──► [HR Manager Approval Queue] ──(Approve)──► [Status: APPROVED] ──► (Syncs Attendance to 'WFH')
└──(Reject)──► [Status: REJECTED] ──► (Notifies Employee)

[HR Manager Submits WFH] ──► [Super Admin Approval Queue]──(Approve)──► [Status: APPROVED] ──► (Conflict of Interest Protected)
└──(Reject)──► [Status: REJECTED] ──► (Notifies HR Manager)
```

09 CONCLUSION & PROJECT EVALUATION

The **VorldX Attendance 360 & OfficeNova Portal** successfully delivers a resilient, cloud-ready, and mobile-native workforce governance ecosystem for VorldX Industries Private Limited. All 18 target screens, 16 HR modules, dynamic cryptographic QR rotating kiosks, and Android APK client applications have been implemented, tested, and validated with zero runtime errors.

10 FUTURE SCOPE & TECHNICAL ROADMAP

1. Edge AI Facial Biometrics:
Integrating on-device lightweight TensorFlow.js face recognition on the mobile app to verify real-time user presence during QR scanning without transmitting sensitive biometric data to cloud servers.

2. Bluetooth Low Energy (BLE) Geofenced Beacons:
Deploying micro BLE beacons inside office physical premises to auto-clock employees upon entering designated office zones, completely eliminating manual check-in requirements.

3. Automated Statutory Tax & Filing Integrations:
Direct API gateways with Indian Employee Provident Fund Organization (EPFO) and Tax Deducted at Source (TDS) portals to automate monthly compliance filings directly from payroll records.

4. Predictive Absenteeism Analytics:
Machine learning algorithms to detect burnout trends, irregular punch patterns, and forecast staffing needs across engineering and customer support departments.

11 PRODUCTION DEPLOYMENT DIRECTORY

Asset	Location / Link	Status
Live Web Application	`https://client-mauve-eight-72.vercel.app`	LIVE PRODUCTION
GitHub Source Code	`https://github.com/shreyay1310-svg/vorldx`	SYNCHRONIZED (MAIN)
Enterprise Database	`/database` (MSSQL Master Setup + SQLite Master Dump)	COMPLETE SCHEMAS
Figma Design System	`/design` (vorldx_tokens.json & FIGMA_DESIGN_SYSTEM.md)	W3C TOKENS

PROJECT VERIFICATION & TECHNICAL ENDORSEMENT

VORLDX INDUSTRIES PRIVATE LIMITED • Lucknow HQ, Uttar Pradesh, India

Approved By: CTO Muba Khan Sir

Chief Technology Officer & Principal Enterprise System Architect

OFFICIALLY APPROVED

Signed: October 2026 • Status: Verified